



SCHOOL BUSINESS AND ECONOMICS

DEPARTMENT ACCOUNTING AND FINANCE

FINANCIAL MODELING

(AcFn3051)

JUNE, 2023

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CHAPTER ONE

INTRODUCTION TO FINANCIAL MODELING AND VALUATION

- Financial modeling is the task of building an abstract representation, called financial models, of a real-world financial situation.
- It is a mathematical model constructed to denote a simplified version of the performance of a financial asset (portfolio of a business), project or any other inv't.
- It is the process of estimating financial performance of a project or business by taking into account all relevant factors, growth and risk assumptions and interpreting their impact.
- It enables the user to acquire a concise knowledge of all the variables involved in Financial Forecasting.
- What is Financial Model used for?
- It can be done for various situations,
- Example:
 - ✓ Valuation of a Company,
 - ✓ Valuation of an Asset,
 - ✓ Pricing Strategies,
 - ✓ Restructuring Situations (Merger & Acquisition), etc.

Financial Modeling is generally used for

- ✓ Risk management
- ✓ Valuation of a Company,
- ✓ Valuation of an Asset,
- ✓ Mergers and acquisitions
- ✓ Option Pricing
- ✓ Budgeting and forecasting
- ✓ Capital allocation
- ✓ Raising capital

Who builds Financial Models?

Investment Bankers

- Equity Research Analysts
- Credit Analysts
- Risk Analysts

- Data Analysts

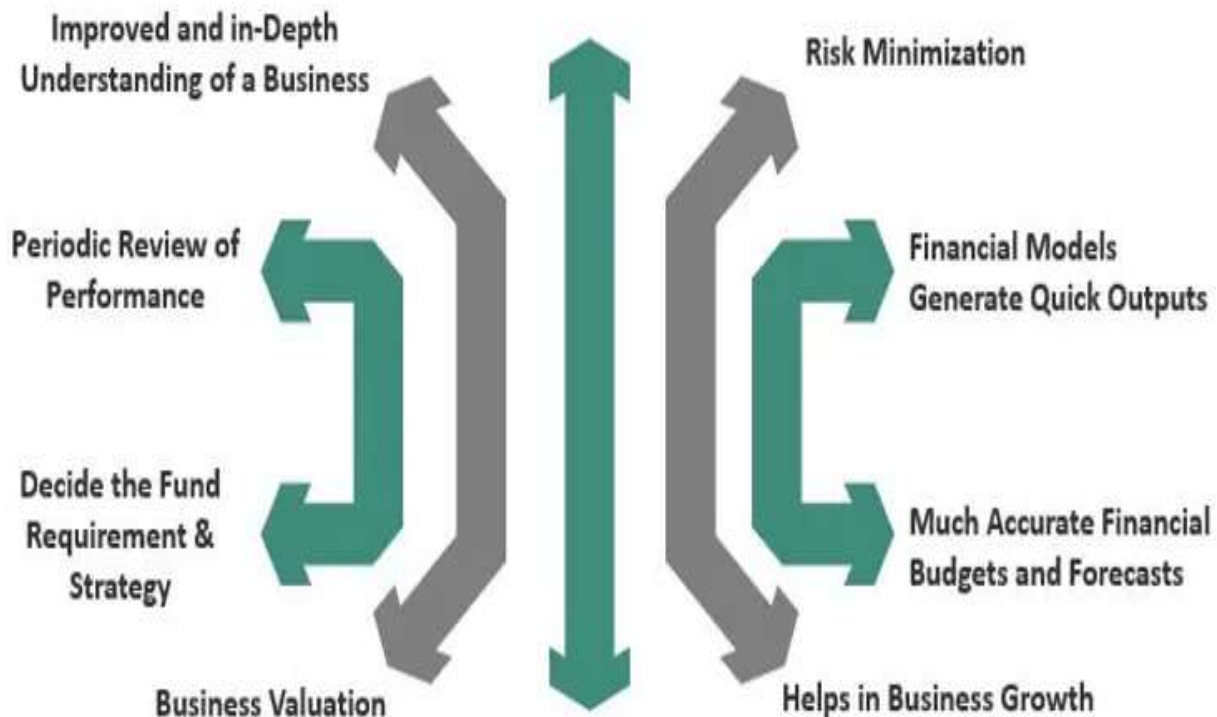
Portfolio Managers

- Investors
- Management/Entrepreneurs
- Majorly modeling is used for determining reasonable Forecasts, Prices for Markets/Products, Asset or **Enterprise Valuation (Discounted CF Analysis, Relative Valuation)**, Share Price of Companies, Synergies, Effects Of Merger/Acquisition on Companies, LBO, Corporate Finance Models, Option Pricing, etc.
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Financial Modeling Examples

- **Financial modeling include four statement – Modeling:**
- ❖ **Discounted cash flow (DCF):** a valuation method used to estimate the value of an inv't based on its expected future cash flows
- ❖ **Initial Public Offering (IPO):** process of offering shares of a private corporation to the public in a new stock issuance.
- ❖ **Leveraged Buyout (LBO):** occurs when the buyer of a company takes on a significant amount of debt as part of the purchase.
- **Buyer will use assets from purchased company as collateral & plan to pay-off debt using future cash flow**
- ❖ **Option Pricing (option premium):** a price buyer of options contract pays for right to buy/sell a security at a specified price in the future
- There are d/t in type and complexity as the situation demands.
- Widely used for **Valuation, Sensitivity Analysis, and Comparative Analysis.**
- There are **other uses**, like **risk prediction, pricing strategy, effects of synergies**, etc.

Financial Modeling Benefits



1) Improved and in-Depth Understanding of a Business

- A **financial model** is developed after having a **deep insight** into the business.
- The analysts understand how a **business operates** and what are **d/t factors that could impact such business**.
- Businesses are also required to understand what changes are expected to take place in a scenario when there are changes in **internal as well as external env'r't of the business**.
- Thus, companies which develop financial models can **understand their business** as well as the **factors affecting them better than their competitors** and are therefore **better prepared for any uncertain situation**.

2) Periodic Review of Performance

- To understand how a business is performing, it is important to do the **variance analysis**
- Financial models help in carrying out the variance analysis by **comparing the actual results of the business against the budgets**.
- **Performance review** can be **done periodically** to get feedback on business operations

- Some advanced financial models help businesses to carry out adjustments in their operations based on variance analysis, so that **overall profits can be improved**.

3) Decide the Fund Requirement & Strategy

- **Financial models provide clarity on the expected cash inflows and outflows.**
- Business can get to **know net CFs** that it would be required to arrange/run its affairs
- The next step is to decide the source of funding, i.e. **debt vs equity**.
- For this, the financial models help in understanding what would be the **cash flow position after meeting the interest expense and repayment of loans**.
- This helps to decide whether and to what extent the business can take debt and what shall be the level of **equity financing**.

4) Business Valuation

- **Companies that wish to know their worth can use f/n models.**
- A financial model helps in **determining free cash flows** that are expected to accrue to a business at d/t points of time which **further helps in reaching fair value of a business**.
- This becomes useful for businesses when making any **restructuring**, such as when **selling the stake** to the outsider parties and investors.

5) Risk Minimization

- Since a financial model helps in carrying out due **diligence by suggesting the financial impact of a particular activity**, thus, **it helps businesses in minimizing the overall risk in a business**.
- For example, suppose a business wants to enter the new market; a financial model would guide the business regarding the cost of such entrance, **effect of marketing, price changes, and so on**.

6) Financial Models Generate Quick Outputs

- Businesses may take months to **get answers to certain financial questions and to determine the impact of a certain decision**.
- However, **financial models are quick in giving results** which helps in **quick decision-making**.
- In such a way, financial models become very useful for businesses.

7) Much Accurate Financial Budgets and Forecasts

- **Financial models build financial budgets and forecasts based on business data and thus, tends to be accurate.**
- Businesses can use these budgets and forecasts for their business activities so that their **activities remain structured and within the defined structure**.
- Not following any budget or business strategy can be harmful to the businesses.



8) Helps in Business Growth

- How the **financial models help a business to grow** is by **suggesting the areas that are capable of generating higher profits.**
- Help in **carrying out a cost-benefit analysis** of new projects.
- **Businesses** can use **financial models to understand** as inv't shall be made in w/c areas & projects for better **profitability and growth.**

Top 10 Types of Financial Models

- There are many different types of financial models.
- In this guide, we will outline top 10 most common models used in corporate finance by financial modeling professionals.

Here is a list of 10 most common types of financial models:

1. Three Statement Model
2. Discounted Cash Flow (DCF) Model
3. Merger Model (M&A)
4. Initial Public Offering (IPO) Model
5. Leveraged Buyout (LBO) Model
6. Sum of the Parts Model
7. Consolidation Model
8. Budget Model
9. Forecasting Model
10. Option Pricing Model

S.No	Financial Model	Description

1	Three Statement Model	Three stmt model is the most basic setup for financial modeling. In this model three stmts (income stmt, B/sheet, and cash flow) are all dynamically linked with formulas in Excel. The objective is to set it up so all the accounts are connected and a set of assumptions can drive changes in the entire model. It's important to know how to link the three f/stmts, which requires a solid foundation of accounting, finance, and Excel skills.
S.No	Financial Model	Description
2	Discounted Cash Flow (DCF) Model	DCF model builds on the 3 stmt model to value a company based on the NPV of the business' future cash flow. DCF model takes the cash flows from the 3 statement model, makes some adjustments where necessary, and then uses the XNPV function in Excel to discount them back to today at the company's Weighted Average Cost of Capital (WACC).

3	Merger (M&A)	Model	M&A model is a more advanced model used to evaluate pro forma accretion/dilution of a merger or acquisition. Common to use a single tab model for each company, where the consolidation of Company A + Company B = Merged Co. The level of complexity can vary widely. This model is most commonly used in inv't banking and/or corporate dev't.
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4	Initial Offering Model	Public (IPO)	Inv't bankers and corporate dev't professionals also build IPO models in Excel to value their business in advance of going public. These models involve looking at comparable company analysis in conjunction with an assumption about how much investors would be willing to pay for the company in question. Valuation in IPO model includes "IPO discount" to ensure the stock trades well in the secondary market.
5	Leveraged Buyout (LBO) Model		A leveraged buyout txn typically requires modeling complicated debt schedules and is an advanced form of financial modeling. LBO is often one of the most detailed and challenging of all types of financial models, as many layers of financing create circular references and require CF waterfalls. These types of models are not very common outside of private equity or inv't banking.

6	Sum of the Parts Model	This type of model is built by taking several DCF models and adding them together. Next, any additional components of the business that might not be suitable for a DCF analysis (e.g., marketable securities, which would be valued based on the market) are added to that value of the business. So, for example, you would sum up (hence “Sum of the Parts”) value of business unit A, business unit B, and inv’ts C, minus liabilities D to arrive at the Net Asset Value for the company.
7	Consolidation Model	This type of model includes multiple business units added into one single model. Typically, each business unit has its own tab, with a consolidation tab that simply sums up the other business units. This is similar to a Sum of the Parts exercise where Division A and Division B are added together and a new, consolidated worksheet is created.
8	Budget Model	This is used to model finance for professionals in Financial Planning & Analysis (FP&A) to get budget together for the coming year(s). Budget models are typically designed based on monthly or quarterly figures and focus heavily on the income statement.
9	Forecasting Model	This type is also used in financial planning and analysis (FP&A) to build a forecast that compares to the budget model. Sometimes the budget and forecast models are one combined workbook and sometimes they are totally separate.

10	Option Pricing Model	Two main types of option pricing models are binomial tree and Black-Scholes. These models are based purely on mathematical formulas rather than subjective criteria and, therefore, are more or less a straightforward calculator built into Excel.
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Financial Modelling in Excel

- Financial modelling in Excel refers to tools used for preparing the expected financial statements predicting the company's financial performance in a future period using the assumptions and historical performance information.

BASIC FINANCIAL CALCULATIONS USING EXCEL

1. FUTURE VALUE (FV)

- FV is **value of a current asset at a future date based on an assumed rate of growth.**
- It is important to investors and financial planners, as they use it to estimate how much investment made today will be worth in the future.

Formula: FV (Rate, Nper, [Pmt], PV, [Type])

The way to calculate in EXCEL

=FV(Rate, Nper, PMT, PV, Type)

- **Rate** = It is the interest rate/period
- **Nper** = Number of periods
- **[Pmt]** = Payment/period
- **PV** = Present Value
- **[Type]** = When the payment is made (if nothing is mentioned, it's assumed that the payment has been made at the end of the period)

Example Calculation:

1. Mr. Alpha has invested the US \$100 in 2016. The payment has been made yearly. The interest rate is 10% p.a. What would be the FV in 2019?
2. Mr. Bilal has invested the US \$12500 in 2014. The payment has been made yearly. The interest rate is 18% p.a. What would be FV in 2020?
3. Mr. Desu has invested the US \$18750 in 2011. The payment has been made yearly. The interest rate is 23% p.a. What would be FV in 2017?

2. FV SCHEDULE

- The Excel FVSCHEDULE function **returns the future value of a single sum based on a schedule of given interest rates.**
- FVSCHEDULE can be used to find the **future value of an investment with a variable or adjustable rate.**
- **To get future value of principal compound interest.**

Formula:

FVSCHEDULE = (Principal, Schedule)

The way to calculate in EXCEL

= FV SCHEDULE(Principal, Schedule)

Principal = Principal is the present value of a particular investment

Schedule = A series of interest rate put together (in case of excel, we will use different boxes and select the range)

CHAPTER TWO

CORPORATE VALUATION

2.1. Methods to Compute Enterprise Value (EV)

- It is an amount that represents the entire cost of company in case some investor intends to acquire 100% of it.
- Calculated by adding market capitalization, preferred stock, outstanding debt, and minority interest together, and then deducting cash and cash equivalents obtained from balance sheet.
- Enterprise Value (EV) is the measure of a company's total value.
- It looks at the entire market value rather than just the equity value, so all ownership interests and asset claims from both debt and equity are included.
- It is an effective cost of buying a company or theoretical price of a target company (before a takeover premium is considered).
- Cash and cash equivalents are deducted from EV since post-acquisition of complete ownership of the company; the cash balance belongs to the new owner.
- Mathematically, it is represented as:
- $EV = \text{Market Capitalization} + \text{Market Value of Debt} - \text{Cash and Equivalents}$
- $EV = \text{Market Capitalization} + \text{Preferred stock} + \text{Outstanding Debt} + \text{Minority Interest} - \text{Cash \&}$

Cash Equivalents

Why is Enterprise Value Used?

- Used for multiple purposes such as EV/EBITDA, EV/EBIT, EV/FCF, or EV/Sales for comparable analysis such as trading comps.
- Other formulas, such as P/E ratio, usually don't take cash and debt into account like EV does.
- Hence, two identical companies that have the same market cap may have two d/t EVs.
- For instance, Company 'A' has \$60 million in market cap, \$20 million in cash, and carries no debt.
- Company B, also has \$60 million in market cap, but has no cash, and carries \$30 million of debt.
- In this simple scenario, we can see that Company A is cheaper to buy b/se it doesn't have any debt to pay off creditors.
- It is very useful in **Mergers & Acquisitions**, especially **with controlling ownership interests**.

When we discuss valuation of a company, we may be referring to any of the ff:

- **Enterprise value:**
- Valuing the company's productive activities.
- **Equity:**
 - Valuing shares of a company, whether for the purpose of buying/selling a single share or valuing all of the equity for purposes of a corporate acquisition.
- **Debt:**
 - Valuing the company's debt. When debt is risky, its value depends on the value of the company that has issued the debt.
- **Other:**
 - We may **want to value other securities** related to the company-for example, the firm's warrants or options, employee stock options, etc.

Four Methods to Compute Enterprise Value (EV)

EV of the firm is the value of the firm's core business activities and forms the basis of most corporate valuation models.

- We distinguish four approaches to computing the EV:
 - I. **Accounting approach to EV** - moves **items on B/sheet** so that all operating items are on the left-hand side of B/sheet and all financial items are on the right-hand side.
 - Although most academics sneer at this approach, it is often a useful starting point for thinking about the enterprise value.
 - II. **Efficient markets approach to EV revalues** - to the extent possible- items on the accounting EV B/sheet at market values.
 - Obvious revaluation is to replace firm's book value of equity with mkt value of equity
 - To the extent that we know the market value of other firm liabilities- debt, pension obligations, etc.-this market value will also replace the book values.
 - III. **Discounted cash flow (DCF) approach** - values of EV as PV of the firm's future anticipated FCFs discounted at weighted average cost of capital (WACC).
 - FCFs can best be thought of as CFs produced by the firm's productive assets-its working capital, fixed assets, goodwill, etc.
 - IV. **EV as PV of Free Cash Flows: DCF "Top Down" Valuation**

Formula:

EV = Mkt Capitalization + Pr. stock + O/standing Debt + Minority Interest – Cash & Cash Equiv'ts

Market Capitalization = Current Price of Share X Outstanding paid up equity shares

Preferred stock = Par value X Outstanding number of paid up equity shares

Outstanding Debt = Bank Loans + Corporate Bonds

Minority Interest = Need to find out

Cash & Cash Equivalents = Cash Balance + Fixed Deposit & Current Account Deposits

Network Capital = Current Asset – Current Liability

Current Asset = Cash + Marketable Securities

Current Liability = Accounts payable + Taxes Payable

Calculations: Class Practice**I. EV Basic Calculations**

1. Let us assume that a company ABC Limited has the ff financial info: Calculate EV.

➤ Shares Outstanding	2,000,000
➤ Current Share Price in	\$ 3
➤ Total Debt in	\$3,000,000
➤ Total Cash	\$ 1,000.000

Solution:

EV = Mkt Capitalization + Pr. stock + O/standing Debt + Minority Interest – Cash & Cash Equiv'ts

$$= (2,000,000 \times 3) + 3,000,000 - 1,000,000$$

$$\begin{aligned}
&= 6,000,000 + 3,000,000 - 1,000,000 \\
&= 9,000,000 - 1,000,000 \\
&= \underline{\underline{\$ 8,000,000}}
\end{aligned}$$

2. Let us assume that a company APPLE Limited has the ff financial infn: Calculate EV

➤ Shares Outstanding	9,500
➤ Current Share Price in	\$ 300
➤ Total Debt in	\$ 1,500,000
➤ Total Cash	\$ 200,000

Solution:

EV = Mkt Capitalization + Pr. stock + O/standing Debt + Minority Interest – Cash & Cash Equiv'ts

$$\begin{aligned}
&= (9,500 \times 300) + 1,500,000 - 200,000 \\
&= 2,850,000 + 1,500,000 - 200,000 \\
&= 4,850,000 - 200,000 \\
&= \underline{\underline{\$ 4,150,000}}
\end{aligned}$$

II. EV through Balance Sheet Model

1. Calculating Apple Inc. EV through Balance sheet Model (in Millions)

➤ Current Market Price	225.74
➤ No of Equity shares	4,754.99
➤ Commercial paper	11,964
➤ Long-term borrowings	102,519
➤ Cash and Equivalent	25,913

Solution:

EV = Mkt Capitalization + Pr. stock + O/standing Debt + Minority Interest – Cash & Cash Equiv'ts

$$= (225.74 \times 4,754.99) + 11,964 + 102,519 - 25,913$$

$$= 1,073,301 + 11,964 + 102,519 - 25,913$$

$$= 1,187,874.44 - 25,913$$

$$= \underline{\underline{\$ 1,161,961.44}}$$

2. Calculating Orange Inc. EV through Balance sheet Model (in Millions)

➤ Current Market Price	47,554
➤ No of Equity shares	6,500
➤ Commercial paper	18,564
➤ Long-term borrowings	175,850
➤ Cash and Equivalent	45,350

Solution:

EV = Mkt Capitalization + Pr. stock + O/standing Debt + Minority Interest – Cash & Cash Equiv'ts

$$= (47554 \times 6,500) + 18,564 + 175,850 - 45,350$$

$$= 309,101,000 + 18,564 + 175,850 - 45,350$$

$$= 3,09,295,414 - 45,3$$

$$= \underline{\underline{\$ 3,09,250.064}}$$

I. Using Accounting Book Values to Value a Company:

- **Firm's Accounting EV (Method of Re-Writing B/Sheet)**
- While we would rarely use accounting numbers to value a company, B/sheet of a company is a useful starting framework for the valuation process.
- In this section we show how accounting stmts can help us define the concept of EV.

Steps to Re-write Balance Sheet

1. Separate **operational versus financial items** in **short-term assets & short-term liabilities**
2. Move **operational current assets** to left side of balance sheet.
3. Move **all debt** (short-term debt, current portion of long-term debt, and long-term) into one debt item.

Steps to be followed

1. Separate operational versus financial items in short-term assets and short-term liabilities.
2. Move operational current assets to the left side of balance sheet.
3. Move all debt (short-term debt, current portion of long-term debt, and long-term) into one debt item.

For Preparing Operational Balance Sheet:

Asset Side

- | | |
|---------------|--|
| Step 1 | Create Liquid Asset = (Cash + Marketable Securities) |
| Step 2 | CA Operational - Remaining Assets |
| Step 3 | Create CL Operational |
| Step 4 | Find Network Capital |
| Step 5 | Remaining Asset Categories |

Liability Side

- | | |
|---------------|---------------------------|
| Step 1 | Find total financial debt |
| Step 2 | Remaining liability Items |
| Step 3 | Find out the Equity |

For Preparing Enterprise Value Balance Sheet:

Step 1 from Financial Debt - Liquid Asset